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Aerosol generating device

Abstract

An aerosol generating device includes: an accommodation portion configured to accommodate an aerosol generating article through an opening; and a deformation guiding portion that comes into contact with one end of the aerosol generating article when the aerosol generating article is inserted into an accommodation portion. The deformation guiding portion deforms the one end of the aerosol generating article when the aerosol generating article is inserted for the first time into the accommodation portion such that the aerosol generating device detects whether an aerosol generating article is being reused based on whether its end is deformed or not.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a National Stage of International Application No. PCT/KR2020/017923 filed Dec. 9, 2020, which claims priority under U.S.C. § 119(a) to Korean Patent Application 10-2020-0001706 filed on Jan. 6, 2020 and Korean Patent Application No. 10-2020-0041537 filed on Apr. 6, 2020.

TECHNICAL FIELD

(2) One or more embodiments relate to an aerosol generating device, and more particularly, to an aerosol generating device including a deformation guiding portion that comes into contact with one end of an aerosol generating article.

BACKGROUND ART

(3) Recently, the need for alternatives to traditional cigarettes has increased. For example, there is growing demand for an aerosol generating device that generates aerosols by heating an aerosol generating material, rather than by combusting cigarettes. Accordingly, studies on a heating-type cigarette and a heating-type aerosol generating device have been actively conducted.

DISCLOSURE

Technical Problem

(4) Users may insert an aerosol generating article into an aerosol generating device to smoke. Generally, an aerosol generating article manufactured for a single use is inserted into an aerosol

generating device, used, and then discarded.

(5) However, some users re-insert a used aerosol generating article without discarding it. The reuse of the aerosol generating article may cause discomfort to the user by changing the flavor of an aerosol generated from the aerosol generating article.

(6) Thus, it is necessary to prevent the reuse of the aerosol generating article.

Technical Solution

(7) One or more embodiments provide an aerosol generating device that deforms at least part of an aerosol generating article that is inserted in the aerosol generating device for the first time, such that reuse of the aerosol generating article is detected and prevented.

(8) Technical solutions to be solved through embodiments are not limited to the above-described technical solutions, and other technical solutions may be inferred from the following embodiments.

Advantageous Effects

(9) In an aerosol generating device according to one or more embodiments, when an aerosol generating article is first inserted, one end of the aerosol generating article is deformed by a deformation guiding portion that comes into contact with one end of the aerosol generating article. As one end of the aerosol generating article is deformed, the aerosol generating device may detect the reuse of the aerosol generating article.

(10) When the reuse of the aerosol generating article is detected, the aerosol generating device may be blocked from operation. Alternatively, the reuse of the aerosol generating article may be notified to a user through an alarm unit installed in the aerosol generating device.

(11) The satisfaction of the user may be prevented from being reduced due to changes in the flavor of an aerosol that may be caused by the reuse of the aerosol generating article.

Description

DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a cross-sectional view schematically showing an aerosol generating device and an aerosol generating article inserted into the aerosol generating device, according to an embodiment.

(2) FIGS. 2A through 2E are cross-sectional views of a deformation guiding portion of an aerosol generating device according to embodiments.

(3) FIGS. 3A, 3B, 3C and 3D are views schematically showing how one end of an aerosol generating article is deformed after the aerosol generating article is first inserted into an aerosol generating device, according to embodiments.

(4) FIG. 4 is a cross-sectional view of an aerosol generating device according to an embodiment.

(5) FIGS. 5A and 5B are cross-sectional views of an aerosol generating device according to another embodiment.

(6) FIGS. 6A and 6B are cross-sectional views of an aerosol generating device according to another embodiment.

BEST MODE

(7) According to an embodiment, an aerosol generating device may include: an accommodation portion configured to accommodate an aerosol generating article through an opening; and a deformation guiding portion configured to come into contact with and deform one end of the aerosol generating article when the aerosol generating article is inserted into the accommodation portion.

(8) The deformation guiding portion may have a shape of part of a sphere protruding toward the opening, a pyramid, a cone, a truncated pyramid, a truncated cone, or a rod having a pointed tip.

(9) A maximum diameter of the deformation guiding portion may be greater than a minimum diameter of a hole formed in the one end of the aerosol generating article, and the deformation guiding portion may be configured to deform the one end of the aerosol generating article by being

inserted into the hole when the aerosol generating article is inserted into the accommodation portion.

(10) The aerosol generating device may further include a sensor configured to detect deformation of the one end of the aerosol generating article.

(11) The sensor may be adjacent to the deformation guiding portion.

(12) The sensor may include: an emission portion configured to emit light or ultrasonic waves toward the one end of the aerosol generating article; and a detection portion configured to receive light or ultrasonic waves reflected from the one end of the aerosol generating article after being emitted from the emission portion.

(13) The sensor may be configured to generate a deformation detection signal based on the deformation of the one end of the aerosol generating article being detected.

(14) The deformation guiding portion may be pressed with a first pressure when an aerosol generating article having no deformed end is inserted, the deformation guiding portion may be pressed by a second pressure when an aerosol generating article having a deformed end is inserted, and the first pressure is greater than the second pressure.

(15) The deformation guiding portion may be configured to generate a deformation detection signal based on being pressed with the second pressure.

(16) The deformation guiding portion may include a pressure sensor configured to measure pressure applied on the deformation guiding portion, and the deformation guiding portion may be configured to generate a deformation detection signal based on the pressure sensor detecting that the deformation guiding portion is pressed with the second pressure.

(17) The aerosol generating device may further include a controller configured to control the aerosol generating device based on the deformation detection signal.

(18) The controller may be configured to block the operation of the aerosol generating device based on receiving the deformation detection signal.

(19) The controller may be configured to generate an alarm signal notifying a user of reuse of the aerosol generating article based on receiving the deformation detection signal.

(20) The aerosol generating device may further include an alarm unit comprising at least one of a vibration unit, a speaker unit, and a display unit, and configured to output at least one of vibration, sound, and an optical signal according to the alarm signal.

MODE FOR INVENTION

(21) With respect to the terms in the various embodiments, the general terms which are currently and widely used are selected in consideration of functions of structural elements in the various embodiments of the present disclosure. However, meanings of the terms can be changed according to intention, a judicial precedence, the appearance of a new technology, and the like. In addition, in certain cases, a term which is not commonly used can be selected. In such a case, the meaning of the term will be described in detail at the corresponding portion in the description of the present disclosure. Therefore, the terms used in the various embodiments of the present disclosure should be defined based on the meanings of the terms and the descriptions provided herein.

(22) In addition, unless explicitly described to the contrary, the word “comprise” and variations such as “comprises” or “comprising” will be understood to imply the inclusion of stated elements but not the exclusion of any other elements. In addition, the terms “-er”, “-or”, and “module” described in the specification mean units for processing at least one function and/or operation and can be implemented by hardware components or software components and combinations thereof.

(23) The terms “embodiment” or “embodiments” described in the specification are intended to explain the inventive concept clearly. Therefore, embodiments should not be construed to be exclusive to each other. For example, elements explained in relating to an embodiment may be embodied and applied in many different forms within the scope of the specification.

(24) In addition, terms used in the present specification are for describing the embodiments and are not intended to limit the embodiments. In the present specification, the singular form also includes

the plurality form unless specifically stated in the phrase.

(25) Throughout the specification, the “longitudinal direction” of a component may be a direction in which the component extends along an axis in one direction of the component, wherein the axis in one direction of the component extends longer than an axis in the other direction of the component crossing the axis in one direction of the component.

(26) As used herein, expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list. For example, the expression, “at least one of a, b, and c,” should be understood as including only a, only b, only c, both a and b, both a and c, both b and c, or all of a, b, and c.

(27) It will be understood that when an element or layer is referred to as being “over,” “above,” “on,” “connected to” or “coupled to” another element or layer, it can be directly over, above, on, connected or coupled to the other element or layer or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly over,” “directly above,” “directly on,” “directly connected to” or “directly coupled to” another element or layer, there are no intervening elements or layers present. Like numerals refer to like elements throughout.

(28) Hereinafter, the present disclosure will now be described more fully with reference to the accompanying drawings, in which exemplary embodiments of the present disclosure are shown such that one of ordinary skill in the art may easily work the present disclosure. The disclosure may, however, be embodied in many different forms and should not be construed as being limited to the embodiments set forth herein.

(29) FIG. 1 is a cross-sectional view schematically showing an aerosol generating device **100** and an aerosol generating article **200** inserted into the aerosol generating device **100**, according to an embodiment.

(30) The aerosol generating device **100** may include an accommodation portion **110** for accommodating the aerosol generating article **200** through an opening and a deformation guiding portion **120** that comes into contact with one end **210** of the aerosol generating article **200** when the aerosol generating article **200** is inserted into the accommodation portion **110**.

(31) The deformation guiding portion **120** of the aerosol generating device **100** may deform one end **210** of the aerosol generating article **200** when the aerosol generating article **200** is inserted into the accommodation portion **110** for the first time.

(32) The aerosol generating article **200** and the accommodation portion **110** may have shapes corresponding to each other. For example, when the aerosol generating article **200** has a cylindrical shape, the accommodation portion **110** may have a cylindrical shape so as to accommodate the aerosol generating article **200**. However, the shapes of the aerosol generating article **200** and the accommodation portion **110** are not limited thereto and may be changed as needed.

(33) The aerosol generating device **100** may include the deformation guiding portion **120**. The deformation guiding portion **120** may be formed on a bottom surface of the accommodation portion **110**, for example. The deformation guiding portion **120** may be a portion extending from the bottom surface of the accommodation portion **110** in a longitudinal direction of the accommodation portion **110**. For example, the deformation guiding portion **120** may be a protrusion protruding from the bottom surface of the accommodation portion **110** by a certain distance.

(34) The deformation guiding portion **120** may include materials having elasticity. When the deformation guiding portion **120** is pressed, a certain portion of the deformation guiding portion **120** may be compressed. The deformation guiding portion **120** may have a rigidity of a certain range or more so as to deform one end **210** of the aerosol generating article **200**.

(35) The deformation guiding portion **120** may come into contact with one end **210** of the aerosol generating article **200** when the aerosol generating article **200** is inserted. When the deformation guiding portion **120** is formed on the bottom surface of the accommodation portion **110**, one end **210** of the aerosol generating article **200** may be a portion facing the bottom surface of the accommodation portion **110** of the aerosol generating article **200**.

(36) When the aerosol generating article **200** is inserted into the accommodation portion **110** for the first time, the deformation guiding portion **120** may come into contact with one end **210** of the aerosol generating article **200** so as to deform one end of the aerosol generating article **200**. That is, as the deformation guiding portion **120** has a rigidity of a certain range or more so to deform one end **210** of the aerosol generating article **200**, one end **210** of the aerosol generating article **200** may receive an external force from the deformation guiding portion **120** when coming contact with the deformation guiding portion **120**. According to the external force from the deformation guiding portion **120**, one end **210** of the aerosol generating article **200** may be deformed.

(37) For example, when the aerosol generating article **200** has a cylindrical shape and one end **210** of the aerosol generating article **200** is a bottom portion of a cylinder, the bottom surface of the cylinder that is one end **210** of the aerosol generating article **200** may come into contact with the deformation guiding portion **120**. The deformation guiding portion **120** may be inserted into the aerosol generating article **200** through the bottom surface of the cylinder that is one end **210** of the aerosol generating article **200**. Insertion of the deformation guiding portion **120** may compress one end **210** of the aerosol generating article **200**. Thus, one end **210** may be deformed.

(38) The cross-sectional area of one end **210** of the aerosol generating article **200** may be greater than that of the deformation guiding portion **120**. Thus, the deformation guiding portion **120** may be completely inserted into one end **210** of the aerosol generating article **200**.

(39) FIGS. 2A through 2E are cross-sectional views showing exemplary aspects of the deformation guiding portion **120** of the aerosol generating device **100** according to an embodiment.

(40) Example shapes of the deformation guiding portion **120** are shown in FIGS. 2A through 2E. The deformation guiding portion **120** may be, for example, a hemispheric shape that is part of a sphere, as shown in FIG. 2A.

(41) As shown in FIG. 2B, the deformation guiding portion **120** may be, for example, a pyramid or a cone extending in the longitudinal direction of the accommodation portion **110**.

(42) The deformation guiding portion **120** may have a shape of a pyramid or a cone including a hollow, as shown in FIG. 2C, for example. In this case, the hollow may have a cylindrical shape, and the configuration of the aerosol generating device **100** or an electric wire for electrical connection may pass through the hollow and extend to the accommodation portion **110**.

(43) The deformation guiding portion **120** may have a truncated pyramid shape or a truncated cone, as shown in FIG. 2D. The truncated cone or pyramid may extend in the longitudinal direction of the accommodation portion **110**. In other words, the deformation guiding portion **120** that comes into contact with the aerosol generating article **200** may have a frustum shape.

(44) The deformation guiding portion **120** may have, for example, a shape of a rod having a pointed tip as shown in FIG. 2E. For example, the deformation guiding portion **120** may have a shape formed by combining a cylinder or prism with a cone or pyramid. For example, a cylinder may extend from the bottom surface of the accommodation portion **110**, and a cone may be combined with the cylinder at the top of the cylinder.

(45) FIGS. 3A, 3B, 3C and 3D are views schematically showing how one end of an aerosol generating article is deformed after the aerosol generating article is inserted into an aerosol generating device for the first time, according to embodiments.

(46) In the example shown in FIGS. 3A and 3B, the aerosol generating article **200** having a hole **215** formed in one end **210** of the aerosol generating article **200** is inserted into the aerosol generating device **100** for the first time.

(47) As shown in FIGS. 3A and 3B, the hole **215** may be formed in one end **210** of the aerosol generating article **200**. The hole **215** may be formed inside the aerosol generating article **200**. The hole **215** may be a space extending from one end **210** of the aerosol generating article **200** toward the other end of the aerosol generating article **200** by a certain distance. The shape and size of the hole **215** may be changed as needed. The hole **215** may have different shapes and sizes according to the type of the aerosol generating article **200**.

(48) When the aerosol generating article **200** having the hole **215** formed in one end **210** is inserted into the aerosol generating device **100** for the first time, as the size of the hole **215** is changed, one end **210** of the aerosol generating article **200** may be deformed. For example, when the aerosol generating article **200** is inserted into the accommodation portion **110**, the deformation guiding portion **120** may be inserted into the through hole **215**. A maximum diameter of the deformation guiding portion **120** may be greater than a minimum diameter of the through hole **215**. The size of the hole **215** may increase according to a difference in diameters of the deformation guiding portion **120** and the hole **215**. One end **210** of the aerosol generating article **200** may be deformed according to the size of the through hole **215**.

(49) FIGS. **3C** and **3D** schematically shows how one end of an aerosol generating article is deformed after the aerosol generating article is inserted into an aerosol generating device for the first time, according to another embodiment.

(50) In the example shown in FIGS. **3C** and **3D**, the aerosol generating article **200** having no through hole in one end **210** is inserted into the aerosol generating device **100** for the first time.

(51) As shown in FIGS. **3C** and **3D**, one end of the aerosol generating article **200** may have a closed shape. When the aerosol generating article **200** is first inserted into the aerosol generating device **100**, the deformation guiding portion **120** may be inserted into the aerosol generating article **200** through one end **210** of the aerosol generating article **200**. Accordingly, a depression may be formed in one end **210** of the aerosol generating article **200**. The depression may be formed to correspond to the shape of the deformation guiding portion **120**. According to the formation of the depression, one end **210** of the aerosol generating article **200** may be deformed.

(52) FIG. **4** is a cross-sectional view of an aerosol generating device **100** according to an embodiment.

(53) The aerosol generating device **100** according to the present embodiment may further include a sensor **130** that detects deformation of one end **210** of the aerosol generating article **200**.

(54) Detection of the initial insertion and reinsertion of the aerosol generating article **200** may be performed in various ways. The aerosol generating device **100** may determine whether the aerosol generating article **200** is inserted, through the deformation guiding portion **120** that comes into contact with the aerosol generating article **200** or an inductance sensor (not shown) mounted on the accommodation portion **110**.

(55) Referring to FIGS. **3A-3B**, when the aerosol generating article **200** is inserted into the accommodation portion **110** for the first time, one end **210** of the aerosol generating article **200** may come into contact with the deformation guiding portion **120** and may be deformed. After one end **210** of the aerosol generating article **200** is deformed according to the initial insertion, when the aerosol generating article **200** is re-inserted into the accommodation portion **110**, the sensor **130** may detect deformation of one end **210** of the aerosol generating article **200**.

(56) As an example, the sensor **130** may measure the height of one end **210** of the aerosol generating article **200**, thereby detecting deformation of one end **210**. When the aerosol generating article **200** is inserted for the first time, a depression may be formed to correspond to the shape of the deformation guiding portion **120**.

(57) The height of the depression may vary according to a distance from the center of the aerosol generating article **200**. For example, the height of the depression may be the greatest at the center of the aerosol generating article **200** and may decrease as the distance from the center of the aerosol generating article **200** increases. When the aerosol generating article **200** is re-inserted into the accommodation portion **110**, the sensor **130** may measure a difference in heights of portions of the depression, thereby detecting deformation of one end **210**.

(58) As another example, the sensor **130** may measure the shape of one end **210** of the aerosol generating article **200**, thereby detecting deformation of one end **210**. The shape of the aerosol generating article **200** may be a preset shape, for example, a cylindrical shape, and one end **210** of the aerosol generating article **200** may correspond to a bottom of a cylinder.

(59) After the aerosol generating article **200** is inserted for the first time, one end **210** of the aerosol generating article **200** may be deformed to have a shape corresponding to the shape of the deformation guiding portion **120**. When the aerosol generating article **200** is re-inserted into the accommodation portion **110**, the sensor **130** may compare the preset shape of one end **210** of the aerosol generating article **200** with the shape of the aerosol generating article **200** inserted into the accommodation portion **110**, thereby detecting deformation of one end **210**.

(60) The sensor **130** may be adjacent to the deformation guiding portion **120**. For example, the sensor **130** may be disposed on the bottom surface of the accommodation portion **110** together with the deformation guiding portion **120**, and a plurality of sensors **130** may be arranged around the deformation guiding portion **120**.

(61) The sensor **130** may include an emission portion **130a** that emits light or ultrasonic waves toward one end **210** of the aerosol generating article **200** and a detection portion **130b** that receives light or ultrasonic waves reflected from one end **210** of the aerosol generating article **200** after being emitted from the emission portion **130a**.

(62) When the aerosol generating article **200** is inserted, the emission portion **130a** of the sensor **130** may emit light or ultrasonic waves toward one end **210** of the aerosol generating article **200**. Alternatively, the emission portion **130a** may periodically emit light or ultrasonic waves toward one end **210** of the aerosol generating article **200**. At this time, a period in which light or ultrasonic waves are emitted, may be preset.

(63) Light or ultrasonic waves emitted from the emission portion **130a** may be reflected from one end **210** of the aerosol generating article **200**. The reflected light or ultrasonic waves may return to the sensor **130**, and the detection portion **130b** of the sensor **130** may receive the reflected light or ultrasonic waves.

(64) The detection portion **130b** that receives the reflected light or ultrasonic waves may detect deformation of one end **210** of the aerosol generating article **200**. The sensor **130** may generate a deformation detection signal when deformation of one end **210** is detected.

(65) The aerosol generating device **100** may further include a battery **160** and a controller **150**.

(66) The battery **160** may supply power used to operate the aerosol generating device **100**. For example, the battery **160** may supply power to heat a heater (not shown) for transferring heat to the aerosol generating article **200**. Also, the battery **160** may supply power required to operate a sensor, a motor, or the like that may be installed in the aerosol generating device **100**.

(67) The controller **150** may control the overall operation of the aerosol generating device **100**. In detail, the controller **150** may control the battery **160**, thereby operating the aerosol generating device **100**. The controller **150** may control the operation of other configurations included in the aerosol generating device **100**. Also, the controller **150** may check the state of each of the configurations of the aerosol generating device **100**, thereby determining whether the aerosol generating device **100** is in an operable state.

(68) The controller **150** may include at least one processor. A processor can be implemented as an array of a plurality of logic gates or can be implemented as a combination of a general-purpose microprocessor and a memory in which a program executable in the microprocessor is stored. In addition, it can be understood by those of ordinary skill in the art that this embodiment may be implemented with other types of hardware.

(69) The controller **150** may be electrically connected to the sensor **130** and may control the overall operation of the aerosol generating device **100**. When the sensor **130** detects deformation of one end **210** of the aerosol generating article **200**, the sensor **130** may transmit a deformation detection signal to the controller **150**.

(70) The deformation detection signal may be generated when the aerosol generating article **200** is re-inserted into the accommodation portion **110**. Once the aerosol generating article **200** is used, one end **210** of the aerosol generating article **200** may be deformed. When the aerosol generating article **200** including the deformed one end **210** is inserted into the accommodation portion **110**,

such deformation of one end **210** may be detected by the sensor **130**.

(71) On detecting deformation of one end **210**, the sensor **130** may generate a deformation detection signal and transmit the deformation detection signal to the controller **150**. When the controller **150** receives the deformation detection signal from the sensor **130**, the controller **150** may detect the reuse of the aerosol generating article **200** and block operation of the aerosol generating device **100**.

(72) As another example, when the sensor **130** does not detect deformation of one end of the aerosol generating article **200**, the sensor **130** may transmit a deformation non-detection signal to the controller **150**. When the deformation non-detection signal is transmitted from the sensor **130**, the controller **150** may recognize the initial use of the aerosol generating article **200** and allow the aerosol generating device **100** to be normally operated.

(73) Transmission of the deformation detection signal to the controller **150** may be one of conditions under which the aerosol generating device **100** may be normally operated. That is, the normal operation of the aerosol generating device **100** may be determined depending on whether the deformation detection signal is transmitted from the sensor **130** to the controller **150**.

(74) As an example, when the deformation detection signal is transmitted from the sensor **130** to the controller **150**, the controller **150** may control power output of the battery **160** to block the normal operation of the aerosol generating device **100**.

(75) As another example, when the deformation detection signal is transmitted from the sensor **130** to the controller **150**, the controller **150** may generate an alarm signal notifying the user of the reuse of the aerosol generating article **200**.

(76) To this end, the aerosol generating device **100** may further include an alarm unit **140**. The alarm unit **140** may include at least one of a vibration unit, a speaker unit, and a display unit, and the alarm unit **140** may be mounted on the aerosol generating device **100** to notify the user of the state of the aerosol generating device **100**. The alarm unit **140** may transmit at least one of vibration, sound, and an optical signal according to the alarm signal of the controller **150**.

(77) For example, the display unit **140** may be mounted as the alarm unit **140** in the aerosol generating device **100**. In this case, as the aerosol generating article **200** including the deformed one end **210** is inserted, the deformation detection signal may be transmitted to the controller **150** is transmitted, and the controller **150** may generate an alarm signal notifying the user of the reuse of the aerosol generating article **200**.

(78) The alarm signal generated by the controller **150** may be transmitted to the display unit that is the alarm unit **140**, and the display unit may display a visible information notifying the user of the reuse of the aerosol generating article **200**. In this case, the visible information may include a certain image or text through which the user may recognize the reuse of the aerosol generating article **200**.

(79) FIGS. 5A and 5B are cross-sectional views of an aerosol generating device **100** according to another embodiment.

(80) A deformation guiding portion **120** of the aerosol generating device **100** according to the present embodiment may distinguish between the initial insertion and re-insertion of the aerosol generating article **200** according to pressure applied on the deformation guiding portion **120**.

(81) When the aerosol generating article **200** is inserted for the first time into the accommodation portion **110**, one end **210** of the aerosol generating article **200** may come into contact with the deformation guiding portion **120** and may be deformed. Deformation of one end **210** of the aerosol generating article **200** may correspond to the shape of the deformation guiding portion **120**. Thus, deformation of one end **210** according to the initial insertion of the aerosol generating article **200** may change the pressure applied one end **210** of the aerosol generating article **200** on the deformation guiding portion **120**.

(82) That is, when the aerosol generating article **200** is inserted for the first time, the deformation guiding portion **120** may be pressed by one end **210** with a first pressure, and when the aerosol

generating article **200** is re-inserted, the deformation guiding portion **120** may be pressed by one end **210** with a second pressure. In this case, the first pressure may be greater than the second pressure.

(83) The first pressure and the second pressure may differ according to the type of the aerosol generating article **200**. In this case, the range of pressures may be predetermined stored as data in the controller **150** of the aerosol generating device **100**.

(84) As shown in FIG. 5B, the deformation guiding portion **120** may be implemented as a membrane switch type and compressed according to external pressure.

(85) In this case, when the deformation guiding portion **120** is pressed with a first pressure, the deformation guiding portion **120** may be compressed equal to or greater than a certain degree. When the deformation guiding portion **120** is compressed equal to or greater than a certain degree, the deformation guiding portion **120** may generate a deformation non-detection signal.

(86) When the deformation guiding portion **120** is pressed with a second pressure that is smaller than the first pressure, the deformation guiding portion **120** may be compressed within a certain range and may generate a deformation detection signal.

(87) For example, when the deformation guiding portion **120** is pressed with the first pressure, the height of the deformation guiding portion **120** may be decreased by a first distance greater than or equal to a threshold, and when the deformation guiding portion **120** is pressed with the second pressure, the height of the deformation guiding portion **120** may be decreased by a second distance less than the threshold. The threshold may be determined by repeated experiments and may differ according to the type of the aerosol generating article **200**.

(88) When the deformation guiding portion **120** is compressed such that its height is decreased by the threshold or more, the deformation guiding portion **120** may generate a deformation non-detection signal. On the other hand, when the deformation guiding portion **120** is compressed such that its height is decreased by less than the threshold, the deformation guiding portion **120** may generate a deformation detection signal. For example, if the deformation guiding portion **120** is not decreased at all (i.e., the second distance is zero), the deformation guiding portion **120** may generate a deformation detection signal.

(89) The deformation detection signal generated from the deformation guiding portion **120** may be transmitted to the controller **150**. When the controller **150** receives the deformation detection signal from the deformation guiding portion **120**, the controller **150** may detect the reuse of the aerosol generating article **200** and block operation of the aerosol generating device **100**.

(90) Generation of the deformation non-detection signal or absence of the deformation detection signal may be one of conditions under which the aerosol generating device **100** may operate normally. That is, the normal operation of the aerosol generating device **100** may be determined according to whether a deformation detection signal is transmitted from the deformation guiding portion **120** to the controller **150**.

(91) As an example, when the deformation detection signal is transmitted from the deformation guiding portion **120** to the controller **150**, the controller **150** may control power output of the battery **160** to block the normal operation of the aerosol generating device **100**.

(92) As another example, when the deformation detection signal is transmitted from the deformation guiding portion **120** to the controller **150**, the controller **150** may generate an alarm signal notifying the user of the reuse of the aerosol generating article **200**.

(93) The alarm signal generation of the controller **150** according to the deformation detection signal and output of the alarm unit **140** according to the alarm signal are described above and thus, a detailed description will be omitted herein.

(94) FIGS. 6A and 6B are cross-sectional views of an aerosol generating device **100** according to another embodiment.

(95) The aerosol generating device **100** according to the present embodiment may include a pressure sensor **127** for measuring pressure. When the pressure applied on the deformation guiding

portion **120** by one end **210** is measured by the pressure sensor **127** as being smaller than a threshold pressure, the deformation guiding portion **120** may generate a deformation detection signal.

(96) In the aerosol generating device **100** according to an embodiment, the pressure sensor **127** may be included in the deformation guiding portion **120**. In this case, the pressure sensor **127** may measure pressure applied on the deformation guiding portion **120** by the deformation guiding portion **120**, and determine whether the measured pressure is smaller than the threshold pressure.

(97) If the measured pressure is greater than or equal to the threshold pressure, the deformation guiding portion **120** may generate a deformation non-detection signal. On the other hand, if the measured pressure is smaller than the threshold pressure, the deformation guiding portion **120** may generate a deformation detection signal.

(98) The deformation detection signal generated by the deformation guiding portion **120** may be transmitted to the controller **150**. When the controller **150** receives the deformation detection signal from the deformation guiding portion **120**, the controller **150** may detect the reuse of the aerosol generating article **200** and block operation of the aerosol generating device **100** or may generate an alarm signal notifying the user of the reuse of the aerosol generating article **200**.

(99) The operation of generating the alarm signal and outputting the alarm unit **140** is described above and thus, a detailed description will be omitted herein.

(100) The aerosol generating device **100** according to one or more embodiments may deform one end **210** of the aerosol generating article **200** through the deformation guiding portion **120** that comes into contact with one end **210** of the aerosol generating article **200** when the aerosol generating article **200** is inserted for the first time. As one end **210** of the aerosol generating article **200** is deformed, the aerosol generating device **100** may detect the reuse of the aerosol generating article **200**.

(101) When the reuse of the aerosol generating article **200** is detected, the operation of the aerosol generating device **100** may be blocked. Alternatively, the reuse of the aerosol generating article **200** may be notified to the user through the alarm unit **40** installed in the aerosol generating device **100**.

(102) Accordingly, the aerosol generating device **100** may prevent the user from unintentionally reusing a single-use aerosol generating article, which causes altered flavor of the aerosol.

(103) At least one of the components, elements, modules or units (collectively “components” in this paragraph) represented by a block in the drawings, such as the controller **150** and the alarm unit **140**, may be embodied as various numbers of hardware, software and/or firmware structures that execute respective functions described above, according to an exemplary embodiment. For example, at least one of these components may use a direct circuit structure, such as a memory, a processor, a logic circuit, a look-up table, etc. that may execute the respective functions through controls of one or more microprocessors or other control apparatuses. Also, at least one of these components may be specifically embodied by a module, a program, or a part of code, which contains one or more executable instructions for performing specified logic functions, and executed by one or more microprocessors or other control apparatuses. Further, at least one of these components may include or may be implemented by a processor such as a central processing unit (CPU) that performs the respective functions, a microprocessor, or the like. Two or more of these components may be combined into one single component which performs all operations or functions of the combined two or more components. Also, at least part of functions of at least one of these components may be performed by another of these components. Further, although a bus is not illustrated in the above block diagrams, communication between the components may be performed through the bus. Functional aspects of the above exemplary embodiments may be implemented in algorithms that execute on one or more processors. Furthermore, the components represented by a block or processing steps may employ any number of related art techniques for electronics configuration, signal processing and/or control, data processing and the like.

(104) Those of ordinary skill in the technical field related to the present embodiments will

appreciate that it may be implemented in a modified form without departing from the essential characteristics of the above-described description. The disclosed methods should be considered in a descriptive sense only and not for purposes of limitation. The scope of the present disclosure is shown in the claims rather than the foregoing description, and all differences within the scope equivalent thereto should be construed as being included in the present disclosure.

INDUSTRIAL APPLICABILITY

(105) One or more embodiments relate to an aerosol generating device including a deformation guiding portion that comes into contact with one end of the aerosol generating article.

Claims

1. An aerosol generating device comprising: a first body configured to accommodate an aerosol generating article through an opening of the first body; at least one second body configured to contact with and physically deform one end of the aerosol generating article, as a direct result of a mechanical interaction of contacting the one end, when the aerosol generating article is inserted into the first body; and a sensor configured to detect, by detecting physical deformation of the one end of the aerosol generating article when the aerosol generating article is re-inserted into the first body, whether the one end of the aerosol generating article has been previously physically deformed by the mechanical interaction by the at least one second body during a prior insertion of the aerosol generating article into the first body.
2. The aerosol generating device of claim 1, wherein the at least one second body has a shape of a protrusion toward the opening.
3. The aerosol generating device of claim 1, wherein the at least one second body has a shape of part of a sphere, a pyramid, a cone, a truncated pyramid, a truncated cone, or a rod having a pointed tip.
4. The aerosol generating device of claim 1, wherein a maximum diameter of the at least one second body is greater than a minimum diameter of a hole formed in the one end of the aerosol generating article, and the at least one second body is configured to deform the one end of the aerosol generating article by being inserted into the hole when the aerosol generating article is inserted into the first body.
5. The aerosol generating device of claim 1, wherein the sensor is adjacent to the at least one second body.
6. The aerosol generating device of claim 1, wherein the sensor comprises: an emission portion configured to emit light or ultrasonic waves toward the one end of the aerosol generating article; and a detection portion configured to receive light or ultrasonic waves reflected from the one end of the aerosol generating article after being emitted from the emission portion.
7. The aerosol generating device of claim 1, wherein the sensor is configured to generate a deformation detection signal based on the deformation of the one end of the aerosol generating article being detected.
8. The aerosol generating device of claim 1, wherein, the at least one second body is configured to be pressed and decreased by a first distance when the aerosol generating article having no deformed end is inserted, the at least one second body is configured to be pressed and decreased by a second distance when the aerosol generating article having a deformed end is inserted, and the first distance is greater than the second distance.
9. The aerosol generating device of claim 8, wherein the at least one second body is configured to generate a deformation detection signal based on being decreased by the second distance.
10. The aerosol generating device of claim 8, wherein the sensor is a pressure sensor that is within the at least one second body, the pressure sensor is configured to measure pressure applied on the at least one second body, and the at least one second body is configured to generate a deformation detection signal based on the pressure sensor detecting that the at least one second body is

decreased by the second distance.

11. The aerosol generating device of claim 7, further comprising a controller configured to control the aerosol generating device based on the deformation detection signal.

12. The aerosol generating device of claim 11, wherein the controller is configured to block operation of the aerosol generating device based on receiving the deformation detection signal.

13. The aerosol generating device of claim 11, wherein the controller is configured to generate an alarm signal notifying a user of reuse of the aerosol generating article based on receiving the deformation detection signal.

14. The aerosol generating device of claim 13, further comprising an alarm unit comprising at least one of a vibration unit, a speaker unit, and a display unit, and configured to output at least one of vibration, sound, and an optical signal according to the alarm signal.

15. The aerosol generating device of claim 1, wherein the sensor is configured to detect the deformation of the one end of the aerosol generating article by measuring: a height of a depression in the one end of the aerosol generating article; or a shape of the one end of the aerosol generating article; or a pressure applied onto the sensor due to re-insertion of the aerosol generating article is into the first body.

16. An aerosol generating device comprising: a first body configured to accommodate an aerosol generating article through an opening of the first body; at least one second body configured to contact with and deform one end of the aerosol generating article when the aerosol generating article is inserted into the first body; and a sensor configured to detect deformation of the one end of the aerosol generating article, wherein the sensor comprises: an emission portion configured to emit light or ultrasonic waves toward the one end of the aerosol generating article; and a detection portion configured to receive light or ultrasonic waves reflected from the one end of the aerosol generating article after being emitted from the emission portion.
